

Design and Evaluation of Deep Learning Models for Continuous Acute Pain Detection Based on Phasic Electrodermal Activity

Javier O. Pinzon-Arenas, *Member, IEEE*, Youngsun Kong , *Member, IEEE*, Ki H. Chon, *Fellow, IEEE*, and Hugo F. Posada-Quintero , *Senior Member, IEEE*

I. INTRODUCTION

Abstract—The current method for assessing pain in clinical practice is subjective and relies on self-reported scales. An objective and accurate method of pain assessment is needed for physicians to prescribe the proper medication dosage, which could reduce addiction to opioids. Hence, many works have used electrodermal activity (EDA) as a suitable signal for detecting pain. Previous studies have used machine learning and deep learning to detect pain responses, but none have used a sequence-to-sequence deep learning approach to continuously detect acute pain from EDA signals, as well as accurate detection of pain onset. In this study, we evaluated deep learning models including 1-dimensional convolutional neural networks (1D-CNN), long short-term memory networks (LSTM), and three hybrid CNN-LSTM architectures for continuous pain detection using phasic EDA features. We used a database consisting of 36 healthy volunteers who underwent pain stimuli induced by a thermal grill. We extracted the phasic component, phasic drivers, and time-frequency spectrum of the phasic EDA (TFS-phEDA), which was found to be the most discerning physiomarker. The best model was a parallel hybrid architecture of a temporal convolutional neural network and a stacked bi-directional and uni-directional LSTM, which obtained a F1-score of 77.8% and was able to correctly detect pain in 15-second signals. The model was evaluated using 37 independent subjects from the BioVid Heat Pain Database and outperformed other approaches in recognizing higher pain levels compared to baseline with an accuracy of 91.5%. The results show the feasibility of continuous pain detection using deep learning and EDA.

Index Terms—Deep learning, electrodermal activity, pain, sequence-to-sequence learning, time-frequency spectrum.

CHRONIC pain is a disease [1] that affects more than 30% of adults in the U.S. [2], [3], representing an economic burden estimated to be more than \$600 billion annually [4]. In clinical practice, the standard for pain sensation assessment is based on highly subjective self-reported scales [5]. There is an increased interest in developing accurate and objective assessment of pain that can contribute to the development of more effective and accurate prescription of pain medication. This is in part because addiction to opioid-based pain medicines has led to a national epidemic of their abuse [6], [7], [8].

Pain affects the autonomic nervous system (ANS), and correlates with a predominant sympathetic activity and diminished parasympathetic activity [9]. Assessing the sympathetic branch is an ideal target for developing biomarkers of pain. The electrodermal activity (EDA) is a simple (does not require complicated circuitry), specific (there is no parasympathetic innervation of sweat glands), and sensitive marker of sympathetic autonomic control obtained from the measurement of changes in conductance of the skin [10], [11], [12]. EDA has been increasingly used to objectively assess pain [13], [14], [15], [16], [17], [18]. Moreover, EDA can be easily implemented in wearable devices. We have previously developed an objective pain level assessment and continuous acute pain detection tools using EDA and machine learning [13], [19], [20], obtaining encouraging results. In this study, we present the results of assessing the feasibility of deep learning techniques such as long short-term memory networks (LSTM) [21] and 1-dimensional convolutional neural networks (1D-CNN) [22], and their fusion, to produce a continuous and accurate biomarker of pain sensation based on EDA.

Conventional machine learning algorithms, namely random forest (RF) and support vector machines (SVM), were used for binary pain (pain vs. no pain) detection using biomedical signals (EDA, ECG, and EMG) [16], [23], [24], [25]. Some studies have used facial video, ECG, EDA, and EMG features independently and combined [25], [26]. Other groups have used only EDA, computing features using time decomposition [16]. Additionally, by fusing of machine learning algorithms with other classifiers, such as kernel density estimation (KDE), and combining EDA and video features, the performance of pain recognition was improved [23]. However, although using video features improved the performance of pain detection in high

Manuscript received 20 October 2022; revised 13 February 2023, 20 March 2023, and 22 May 2023; accepted 30 June 2023. Date of publication 3 July 2023; date of current version 6 September 2023. (Corresponding author: Hugo F. Posada-Quintero.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by University of Connecticut Institutional Review Board, protocol H19-071, and performed in line with the Declaration of Helsinki.

Javier O. Pinzon-Arenas is with the Faculty of Engineering, Universidad Militar Nueva Granada, Bogota 11 #101-80, Colombia (e-mail: est.javier.pinzon@unimilitar.edu.co).

Youngsun Kong, Ki H. Chon, and Hugo F. Posada-Quintero are with the Department of Biomedical Engineering, University of Connecticut, Storrs, CT 06269 USA (e-mail: youngsun.kong@uconn.edu; kchon@engr.uconn.edu; hugo.posada-quintero@uconn.edu).

Digital Object Identifier 10.1109/JBHI.2023.3291955

TABLE I
SUMMARY OF RELATED WORKS

Author	Classifier	Signal	Dataset	Performance measure	Metric	Score
Werner et al. [26]	RF	EDA, ECG, EMG, Video	BioVid Heat Pain Database	10-Fold Cross-Validation	Accuracy	65.7% (49.6, 80.6)
Susam et al. [16]	SVM	EDA	Custom Database	Leave-One-Subject-Out Validation	Accuracy	77.7%
Susam et al. [23]	SVM-KDE	EDA, Video	Custom Database	Leave-One-Subject-Out Validation	Accuracy	91%
Aqajari et al. [24]	RF, KNN	EDA	UCI iHurt Dataset	Leave-One-Subject-Out Validation	Accuracy	71.8% (86.0, 61.5)
Kasaeyan Naeini et al. [25]	SVM, RF	EDA, ECG, EMG, Video	BioVid Heat Pain Database	10-Fold Cross-Validation	Accuracy (only bio-signals) (BL vs T1)	79%
Wang et al. [27]	Hybrid RNN-ANN	EDA, ECG, EMG	BioVid Heat Pain Database	Leave-One-Subject-Out Validation	Accuracy	70.3% (58.5, 83.3)
Lopez-Martinez et al. [28]	MT-NN	EDA, ECG	BioVid Heat Pain Database	10-Fold Cross-Validation	Accuracy	65.8% (51.7, 82.6)
Pouromran et al. [32]	BiLSTM-XGB	EDA	Custom Database	Custom Split	F1-Score	81%
Subramaniam et al. [29]	Hybrid CNN-LSTM	EDA, ECG	BioVid Heat Pain Database	Custom Split	Accuracy	91.4% (93.9, 94.1)
			Custom Database		Accuracy, MSE (Only EDA)	70%, 0.3
Gouverneur et al. [30]	(MLP, CNN, ConvLSTM, CAE) + RF	EDA	BioVid Heat Pain Database	Leave-One-Subject-Out Validation	Accuracy, F1-Score (MLP+RF)	70.9% (59.1, 84.2); 69.7% (57.2, 83.6)
			PainMonit Database		Accuracy, F1-Score (CNN+RF)	70.4% (58.1, 87.4); 69.1% (56.4, 86.7)
Rodriguez et al. [31]	CNN, Hybrid CNN-LSTM	Video	UNBC-McMaster Shoulder Pain Expression Archive Database	Leave-One-Subject-Out Validation	AUC	93.3
Othman et al. [34]	RF, LSTM	EDA, ECG, EMG, Audio, Video	X-ITE Pain Database	Custom Split	F1-Score	63.3%
					RMSE	0.07

pain intensity, the performance drastically decreased for low and medium pain detection [26]. This is possibly linked to the discrepancies between manual and automatic labeling of facial features [23], which heavily depends on external factors, such as illumination, rotation of the face, etc. When no video features are used, pain recognition (even when low pain intensity is presented) better performance is achieved [25], even using only EDA [24].

Deep learning techniques have been applied in binary pain detection, showing better performance compared to machine learning [27], [28], [29], [30], [31], [32]. Multitask networks (MT-NN) [28] showed better results in detecting high-intensity pain, compared with other machine learning techniques using EDA and ECG features [26]. However, this method was implemented in a subject-specific manner—the model was trained using data from the specific subject. Other group used the recurrent neural network (RNN) to learn ECG, EMG, and EDA temporal characteristics [27]. This model achieved a higher overall accuracy than previous machine learning methods implemented in the BioVid database. In [30], they made a comparison of different methods to extract features from EDA signals, using fused deep learning algorithms and random forest (RF) for classification. Remarkably, they obtained better performance using only EDA than other studies using multiple signals. More advanced networks, such as LSTM and CNN have been used. In a study [32], a stacked unidirectional and bidirectional LSTM

concatenated to an extreme gradient boosting decision trees model was used to detect pain in cropped segments of the signal, using the raw EDA signal and its phasic and tonic components as input, obtaining an average F1-Score of 81%. In another study [29], a hybrid CNN-LSTM model obtained an accuracy of 91.4% when the EDA and ECG features were used together. When only EDA was used, the accuracy was 80.3%. In [31], they used a hybrid CNN-LSTM architecture, reporting that temporal features are key for pain detection, showing that combining spatial and temporal features improved the recognition results. However, facial expressions can be easily feigned [33]. A work that examined continuous monitoring of pain was presented in [34]. In this work, the performance of LSTM and RF was tested, using multiples signal, and it was concluded that LSTM had promising results for the continuous classification of stimulation intensity, but lacking a good onset detection.

Table I gives a summary of the works mentioned before. The table shows the approach used to perform the pain detection task and their best scores. When reporting results on the BioVid database, which contains four levels of pain stimulation applied using heat (T1 to T4) according to a baseline temperature (BL), obtaining four pain levels (PA1 to PA4), we included the mean accuracy for the four pain levels (when available), along with the accuracy obtained for classification of PA1 and PA4 versus baseline, using the format “Mean Score PA1-PA4 (BL vs. PA1, BL vs. PA4)”.

Traditional pain assessment methods based on signals require the use of a fixed-length window in which a model is run to detect pain. We hypothesize that window-based pain detection approaches do not provide accurate pain detection because the reaction to pain depends on the physiological and environmental conditions (e.g., stress, relaxation, anticipation, anxiety) prior to the pain stimulus being delivered, and such information is relevant for its detection. Moreover, in many contexts continuous pain monitoring is necessary for allowing immediate intervention [34]. In this study, we trained several architectures that incorporated information from the time series prior to pain, including LSTM with uni- and bi-directional layers, 1D-CNN, and temporal convolutional networks, using different features from EDA signals acquired from healthy volunteers who underwent painful stimulation by means of thermal grills. The main contributions of this article are:

- We have found that the time-frequency spectrum of the phasic EDA signal is a sensitive feature for continuous pain detection using deep learning.
- We have tested the feasibility of sequence-to-sequence detection of pain using EDA and deep learning models, suppressing the need for fixed windows, and providing a precise detection of the onset and end time points of the physiological reaction to pain.

II. MATERIALS AND METHODS

We conducted a systematic evaluation of deep learning techniques and features derived from the EDA signal (such as the phasic component, phasic driver, and the estimated power spectral density) to detect pain. First, we trained single LSTM and CNN network variants, as well as hybrid architectures (serial, residual, and parallel) fusing the traditional deep learning techniques, combining the EDA features to determine the best model in the overall. Later, we evaluated the performance of each model, in addition to the performance achieved using each EDA feature combination for all networks. This systematic analysis is intended to obtain better insight into the most appropriate EDA feature as well as the most suitable type of deep learning network architecture for pain detection. In this section we describe the datasets as well as the models of single networks that we considered in the analysis, the combination architectures, and the EDA features used.

A. ChonLab Thermal Grill Pain Database

The ChonLab Thermal Grill Pain Database was constructed by consolidating two datasets previously published [13], [35]. Consolidated database consisting of thirty-eight healthy volunteers (17 males and 21 females) who underwent pain stimuli induced by a thermal grill. The thermal grill was made with parallel aluminum tubes. Warm (50–58 °C) and cold (4 °C) water flowed in an alternating way, to induce the sensation of pain due to contrasting temperatures. Each subject had to place their right hand for at least five seconds on the thermal grill, which was set to randomly induce two pain levels, low and high.

All EDA signals were downsampled to 4 Hz. Two subjects were discarded due to their low and inconsistent pain responses. A total of thirty-six ($N = 36$) subjects were included in the analysis. All signals were used raw, i.e., despite most of the signals being only partially clean, the artifacts or stress-related activity presented in some of them have not been removed.

1) *Preprocessing of the EDA Signal*: For decomposing the EDA signals into phasic (rapid) and tonic (slow) components, in this study we used the convex optimization approach (cvxEDA) [36]. Since it has been reported that the phasic component contains the most relevant information for acute pain detection [13], we used the phasic EDA as one of the input features, discarding the use of the tonic component. Additionally, we obtained a time-frequency spectrum of the phasic EDA (TFS- phEDA) by means of the Welch's power spectral density method [37]. To calculate the power at each time step, we used a sliding window of 2.5 seconds length (ten samples) and 25 discrete Fourier Transform points to compute the modified periodogram in each window. In total, 17 spectral coefficients, from frequencies 0 Hz to 2 Hz, were obtained. We also used the phasic drivers obtained from the phasic EDA [36]. An example of the obtained EDA signal and its components, the phasic EDA, the TFS-phEDA of the complete signal, and the onset of each stimulus are shown in Fig. 1.

2) *Pain Labeling*: To perform continuous pain detection with time occurrences of pain, the networks were trained to carry out a sequence-to-sequence task, labeling every sample of the signals as “pain” or “no pain.” The phasic drivers were used to determine when the physiological activation caused by pain sensation had begun, after each stimulus was applied. If the difference between the current and next phasic drivers is higher than 0.1, and the slope of the phasic component at the same instant is positive, then that time step is selected as the start of the pain sensation. This threshold was defined by visually exploring the phasic drivers occurring right after pain stimulation was applied. If these conditions are not met in a window of 5 seconds after each stimulus, we assumed that there was no physiological activation. This 5-sec window was set considering that the duration of a skin conductance response (SCR) varies between 1 to 5 seconds, and that previous studies successfully used this window size [13], [38]. To determine the end of the pain sensation manifestation on EDA, the maximum value of the phasic EDA was taken within a 10-sec window after the established onset of the sensation. Note that this window size allowed us to take the complete sustained reaction to pain, even if the subject exhibited a second SCR (a common phenomenon for pain reaction). Then, the maximum value from the two peaks was selected. We used the maximum value because we assumed that when the phasic EDA starts to decrease, the sympathetic activation has ended, and the subject's sympathetic reaction is going back to its previous stage. The bottom plot of Fig. 1 shows the labelling of a given signal. Note that the described windows were used for labeling purposes only, not for training or testing of the networks. In addition, if there was a response prior to a given stimulus' onset, it was labeled as “no pain”, since it may have been caused by anticipation reflexes (e.g., the rise in EDA before the second stimulus in Fig. 1).

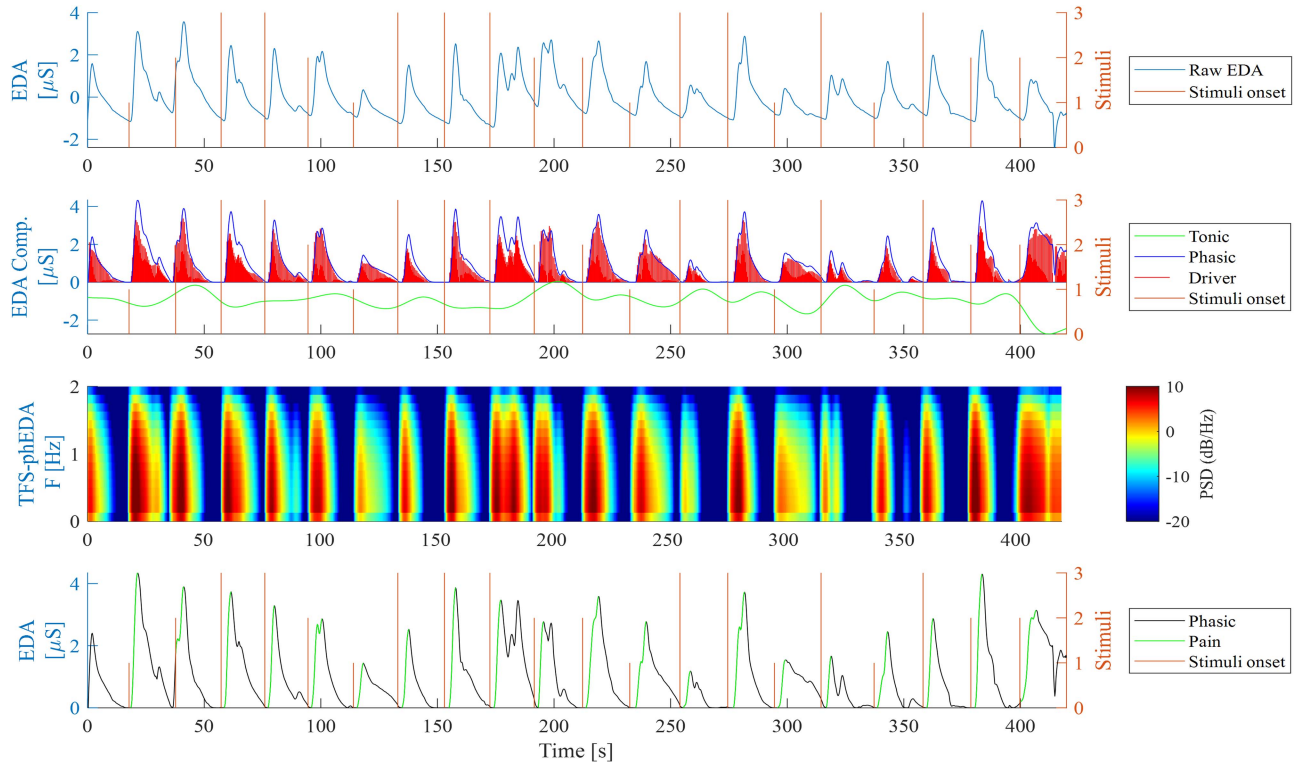


Fig. 1. First panel: Sample raw EDA signal and stimuli onset. Second panel: Tonic and phasic components of EDA, and phasic drivers obtained using *cvxEDA*. Third panel: TFS-phEDA. Fourth panel: pain labelling over the phasic components, in which green sections were labeled as “pain,” and black sections are “no pain.”

3) Experimental Dataset and Subsets Structure: We randomly distributed the thirty-six subjects into three groups. Thirty subjects were used for training, three for validation, and three for test. Moreover, for the purpose of verifying the performance of the classifiers with different inputs extracted from the EDA signals, six subsets were independently tested by using individually or combining the phasic component (Ph), the phasic drivers (dr) and the power coefficients of the TFS-phEDA: 1, Ph only; 2, Ph and dr (Ph,dr); 3, Ph,dr,TFS-phEDA; 4, Ph,TFS-phEDA; 5, dr,TFS-phEDA; and 6, TFS-phEDA only.

B. BioVid Heat Pain Database

To allow direct comparison with previous works, we used the EDA signal of Part C of the BioVid Heat Pain Database [39] as an independent test dataset, which contained the long recording sequences from eighty-seven healthy subjects. The pain data were collected by inducing a pain stimulus via thermode over the right arm. Four equally distributed levels of heat stimulus (T1 to T4, T1 being the temperature where the subject started feeling pain—PA1, and T4 the temperature where the maximum acceptable pain—PA4—was perceived by the subject) were applied for 4 seconds, 20 times for each temperature, with a randomized recovery time of 8–12 seconds. Subjects with low pain response to low intensities were discarded from the database. For the pain labelling, a 3-second shift backwards of the original pain event label given in the database was performed, to give a more accurate pain response onset label. The procedures for

preprocessing and labelling of the EDA signal were performed in the same way as previously described for our dataset.

C. Classifiers

We implemented two variants of the LSTM network, the uni- and bi-directional networks, that have been widely used for sequential tasks. On the other hand, CNN has demonstrated its suitability and versatility in time-series processing, being able to process 1-dimensional data inputs rather than the conventional 2-dimensional inputs [22]. We also tested the temporal convolutional network (TCN), that is a kind of CNN architecture, which has been recently developed and has shown promising results for processing time series [40]. The output of all architectures consisted of a fully connected layer with 2 neurons, referring to the classes (Pain/No Pain), a SoftMax and classification layer. More details on each architecture tested in this study are provided below.

1) LSTM Architectures: The LSTM network is a special type of recurrent network that was intended to overcome the long-term dependencies of large time sequences [21]. The LSTM can be uni- (uniLSTM) or bi-directional (biLSTM) [41]. The main difference between the uniLSTM and the biLSTM networks is that the former keeps information from the past, incorporating past information about the signal to better identify the occurrence of an event (pain, in this study), while the latter allows the network to use information from both the past and future samples, since the information flows in both directions [42].

TABLE II
PARAMETERS FOR EACH LSTM AND CNN ARCHITECTURE

Network	Parameter	Value
uniLSTM	Layer 1	100, 200
biLSTM	Layer 2	100, 200
SBU-LSTM	Layers	2, 3, 4
1D-CNN	Filters	32, 64, 128
	Filter Size	5
TCN	Blocks	2, 4, 6, 8
	Filters	128
	Filter Size	3, 5, 12
	Output (2 classes)	Fully connected layer SoftMax Classification
	Solver	Adam

Building a deeper architecture by stacking layers increases the complexity of the network, potentially allowing it to learn higher representations of the signal and improving its performance [43], [44]. For this reason, we systematically evaluated 1-layer and 2-layer architectures. Additionally, it has been reported that mixing uniLSTM and biLSTM (called stacked bi-directional and uni-directional LSTM or SBU-LSTM architecture [45]) may enhance the performance of the network [43]. We used a biLSTM as the first layer, to enable the network to learn the temporal and spatial representations of the signal when pain is detected, while considering what happens to the signal when the reaction ends. We used a uniLSTM as the second layer, to allow the network to learn the forward dependencies from the features provided by the first layer. For the systematic evaluation of the uniLSTM and biLSTM networks of 1 and 2 layers, as well as of the SBU-LSTM architecture, we tested combinations of 100 and 200 hidden units of each layer in each architecture. Table II summarizes the models and parameters used.

2) CNN Architectures: Although CNNs are used mostly in 2-dimensional approaches [46], after Kiranyaz et al. introduced a model using 1-dimensional filters for electrocardiogram signal classification [47], the CNN started to gain popularity in applications that require it to analyze sequence tasks involving signal processing [22]. We also used a variant of 1D-CNN, termed temporal convolutional network (TCN), developed by Bai et al. [40]. The TCNs are 1D-CNNs that use “dilated causal convolutions” and residual blocks. The “casual” term means that a convolution operation is applied from time “ t ” backwards, and the “dilated” term refers to an exponential increment of the convolution filter size as the depth of the network increases, skipping intermediate values to cover a larger section of the sequence, thus increasing its receptive field.

Considering the abilities of the 1D-CNNs, we have included the conventional 1D-CNN with causal filters and TCN in the systematic comparison with the LSTM networks explained above for the task of sequence-to-sequence acute pain detection. For 1D-CNN, we used different values for the depth (number of layers) and the number of filters used in each layer (each two layers, the initial value is doubled), maintaining a filter size of 5. In the TCN, we used different values for the depth (number of

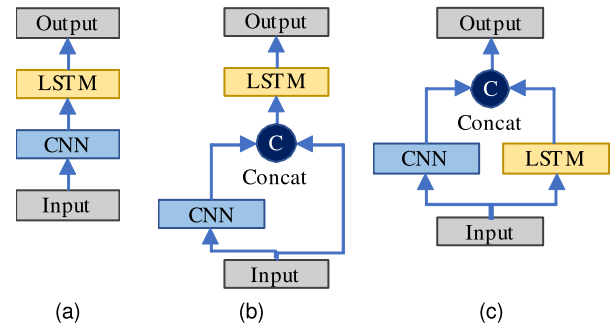


Fig. 2. Hybrid architectures proposed. (a) Series network. (b) Residual network. (c) Parallel network. The input corresponds to the features of the entire EDA signal, and the output is the pain/no pain classification at each signal sample.

residual blocks) and the size of the filter, maintaining the number of filters for all the convolutional layers. Additionally, we used Layer Normalization [48] and spatial dropout (randomly turning off entire channels with a dropout factor of 0.005) between convolutional layers. Each residual block contains two sets containing a dilated casual convolution – layer normalization – ReLU, ending with the spatial dropout. Table II summarizes the models and parameters used.

3) Hybrid CNN-LSTM Architectures: In addition, we implemented CNN-LSTM hybrid architectures of three types. The first one (Fig. 2(a)) is a conventional CNN-LSTM connected in series. In this architecture, the function of the CNN is to extract the spatial features of the sequence, and then feed the LSTM with those features to learn their temporal dynamics [29]. However, depending on the internal structure of the CNN, some original features of the signal may be lost at the time they are given to the LSTM. For this reason, the second architecture was considered (Fig. 2(b)) for which we added a residual connection directly from the input concatenated with the output features of the CNN. Doing this, the LSTM is fed with both the original input features and the spatial features extracted by the CNN. For the third architecture, we implemented a parallel structure (Fig. 2(c)) of a CNN and a LSTM without shared connections. The main idea of this architecture is to let the network learn both spatial and temporal features independently [49], [50], [51] and, after a concatenation of those features, they are fed to the output layer of the network.

Based on the best performance of the CNN and LSTM networks when tested individually (Section III), the parameters of those best models were used for the hybrid architectures. Considering this, for the CNN section of the hybrid architecture, we used the 3-layer 1D-CNN models with 64 and 128 filters in the first layer, as shown in Table III. For the TCN, we used 6 and 8 blocks with filter size of 5. For the LSTM section, the 2-layer uniLSTM, 2-layer biLSTM, and SBU-LSTM with 200 hidden units were used. Additionally, the shallowest version of that combination (a 3-layer 1D-CNN, 64 initial filters, with a 2-layer biLSTM with 100 hidden units) were implemented. The output section of the hybrid architectures is the same as the single ones. Table III shows a summary of the hybrid architectures. We labeled the models using the scheme “Type of architecture +

TABLE III
PARAMETERS FOR EACH HYBRID CNN-LSTM ARCHITECTURE

Type	CNN model	LSTM model	# Filters - # Blocks	Hidden Units
Series Residual Parallel	1DCNN	uniLSTM	64 x2 / 128	100
		uniLSTM	64 x2 / 128	200
		biLSTM	128 x2 / 256	
Series Residual Parallel	TCN	SBU-LSTM		
		uniLSTM	6	200
		biLSTM	8	
		SBU-LSTM		

CNN model + LSTM model.” For instance, “Series 1DCNN-biLSTM” corresponds to a Series architecture using a 1D-CNN first followed by a biLSTM.

D. Training Parameters and Model Performance Evaluation

To provide a fair comparison between all models, each network was trained with the same parameters: 300 training epochs with an initial learning rate of 0.001 and a drop factor of 0.1 applied at epoch 250, a batch size of 1 for training networks that included LSTM only, and a batch size of 10 for the CNN and hybrid models. Moreover, the same training, validation and test datasets and the minimum validation loss were used to select the higher performance obtained during the training.

The dataset was imbalanced. We obtained 1932.8 ± 981.5 and 309.6 ± 101.1 samples per subject for “No pain” and “Pain” categories in the training set, respectively (mean $\pm$ standard deviation). This imbalance can cause the networks to be biased towards the class with more samples in the database during the training. To deal with class imbalance, we performed a weight balance by means of inverse class frequency weighting. This is a widely used tool to deal with class imbalance [52]. We used the harmonic mean of the precision and sensitivity, also known as balanced F-Score (F1-score), as the main performance measurement, using the “Pain” class as target. In the same way, as support parameters, the sensitivity, specificity, global accuracy, and the area under the receiver operating characteristic curve (AUC) of each network were computed.

III. RESULTS

In total, 75 architectures were implemented and each one was trained with the six subsets containing a combination between the phasic EDA, phasic driver, and TFS-phEDA, as mentioned in Section II-A, for classifying “pain” vs “no pain” in each sample of the EDA signal. In total, we trained 450 models. Although all the signals had different lengths, training and testing were carried out by feeding the entire signals to the networks; i.e., no cropped signals or fixed windows were used to feed the models.

A. Evaluation of the Models

The networks with the best performance according to F1-score are shown in Table IV, along with the performance evaluation metrics for the analysis. For each metric, the best model

TABLE IV
TOP 5 OF THE BEST PERFORMANCE OBTAINED BASED ON F1-SCORE

Network	Input	Top 5 Global F1-Score				
		F1-S	Acc	Sens	Spec	AUC
Parallel TCN-SBU-LSTM	dr, TFS-phEDA	77.8	93.1	82.3	95.0	0.967
Parallel TCN-biLSTM	Ph, TFS-phEDA	77.0	93.7	72.5	97.3	0.954
Parallel TCN-SBU-LSTM	TFS-phEDA	76.9	92.9	80.1	95.2	0.959
Parallel 1DCNN-SBU-LSTM	Ph, TFS-phEDA	76.9	92.7	82.5	94.5	0.961
Parallel TCN-uniLSTM	TFS-phEDA	76.5	93.5	72.8	97.0	0.940

F1-S: F1-score; Acc: Accuracy; Sens: Sensitivity; Spec: Specificity; AUC: Area under the ROC curve.

TABLE V
COMPARISON BETWEEN THE BEST GLOBAL F1-SCORE OF EACH NETWORK TYPE WITH THE RESPECTIVE INPUT USED

Network	Input	F1-S	Acc	Sens	Spec	AUC
uniLSTM	Ph, TFS-phEDA	73.6	91.7	79.0	93.9	0.939
biLSTM	Ph, TFS-phEDA	74.9	92.0	81.2	93.9	0.952
SBU-LSTM	Ph, dr, TFS-phEDA	75.3	92.5	78.4	94.9	0.949
TCN	Ph, dr, TFS-phEDA	70.2	91.3	69.7	95.0	0.930
1DCNN	dr, TFS-phEDA	68.0	89.7	74.2	92.4	0.913
Series 1DCNN-SBU-LSTM	Ph	72.2	92.2	69.2	96.1	0.936
Residual 1DCNN-SBU-LSTM	Ph, dr, TFS-phEDA	74.6	92.4	76.4	95.1	0.949
Parallel 1DCNN-biLSTM	Ph, TFS-phEDA	76.9	92.7	82.5	94.5	0.961
Series TCN-SBU-LSTM	TFS-phEDA	75.7	92.6	78.1	95.1	0.945
Residual TCN-SBU-LSTM	Ph	75.6	92.8	75.5	95.8	0.941
Parallel TCN-SBU-LSTM	dr, TFS-phEDA	77.8	93.1	82.3	95.0	0.967

is highlighted in bold. Due to the diverse length and number of stimuli performed on the subjects as well as having more subjects for the analysis, we merged validation and test subsets to evaluate the global performance of all networks.

The ten highest F1-scores were obtained by the hybrid parallel models using as part of their input the TFS-phEDA. The best model was the Parallel TCN-SBU-LSTM when using the phasic driver and the TFS-phEDA as input features, obtaining an F1-score of 77.8%. The best model based on F1-score also exhibited high overall accuracy (not the highest), sensitivity, and AUC. In addition to F1-score evaluation, based on ROC curves, parallel TCN-SBU-LSTM was the most suitable model for classifying pain from no pain samples. Given the discrepancy in performance for the validation and test, we are reporting F1-score for both datasets. Table V shows the best results for each type of architecture along with the phasic EDA features

TABLE VI
COMPARISON OF THE BEST F1-SCORE OBTAINED DEPENDING
ON THE INPUT

Network	Input	F1-S	Acc	Sens	Spec	AUC
Residual TCN-SBU-LSTM	Ph	75.6	92.8	75.5	95.8	0.941
Parallel IDCNN-biLSTM	Ph, dr	75.5	92.6	77.5	95.2	0.962
Parallel TCN-SBU-LSTM	Ph, dr, TFS-phEDA	75.5	92.7	76.6	95.5	0.953
Parallel TCN-biLSTM	Ph, TFS-phEDA	77.0	93.7	72.5	97.3	0.954
Parallel TCN-SBU-LSTM	dr, TFS-phEDA	77.8	93.1	82.3	95.0	0.967
Parallel TCN-SBU-LSTM	TFS-phEDA	76.9	92.9	80.1	95.2	0.959

which it was trained on. Most models achieved the maximum performance when using the TFS-phEDA along with another feature such as the phasic component or the phasic driver.

Table VI shows the models with the highest F1-scores with each phasic EDA feature considered. As mentioned before, hybrid parallel architectures have shown greater results with the different combination of EDA features for sequence-to-sequence pain recognition. Additionally, using either TFS-phEDA alone or adding another EDA component outperformed the other combinations, increasing the F1-score $\sim 1\%$. Moreover, using dr,TFS-phEDA or only TFS-phEDA, the models obtained more than 80% sensitivity.

B. Performance Analysis of the TCN-SBU-LSTM Model

We observed that the network successfully recognized pain presence in some subjects with an F1-score, sensitivity, and specificity of around 95% (Fig. 3). In those cases, the network was precise at classifying the start and end of the pain activation, only making a few mistakes such as in the small peak near second 250 (Fig. 3(a)). Also, in some challenging signals, the network discarded those activations that were not related to pain. Fig. 3(b) shows a signal with EDA variations that were related to stress and anxiety caused by pain anticipation, but not actual pain. Visible reactions before pain stimulation were deemed to be produced by stress and anxiety. In those cases, the model was able to recognize most of the pain activations. However, many large non-pain related SCRs were misclassified as a manifestation of pain. Additionally, due to the small activity level presented in some pain activations, such as the one around 300 seconds, the network mislabeled such SCR as “no pain”. In this case, the network obtained an F1-score of 73.4%, sensitivity of 79.7%, and specificity of 85.2%.

C. BioVid Database Test

To further examine the robustness of the TCN-SBU-LSTM model, we used the network on thirty-seven subjects from the BioVid Heat Pain Database. The network was applied over the entire signals, obtaining a label at each signal time step. To evaluate the performance of the detection at each pain level,

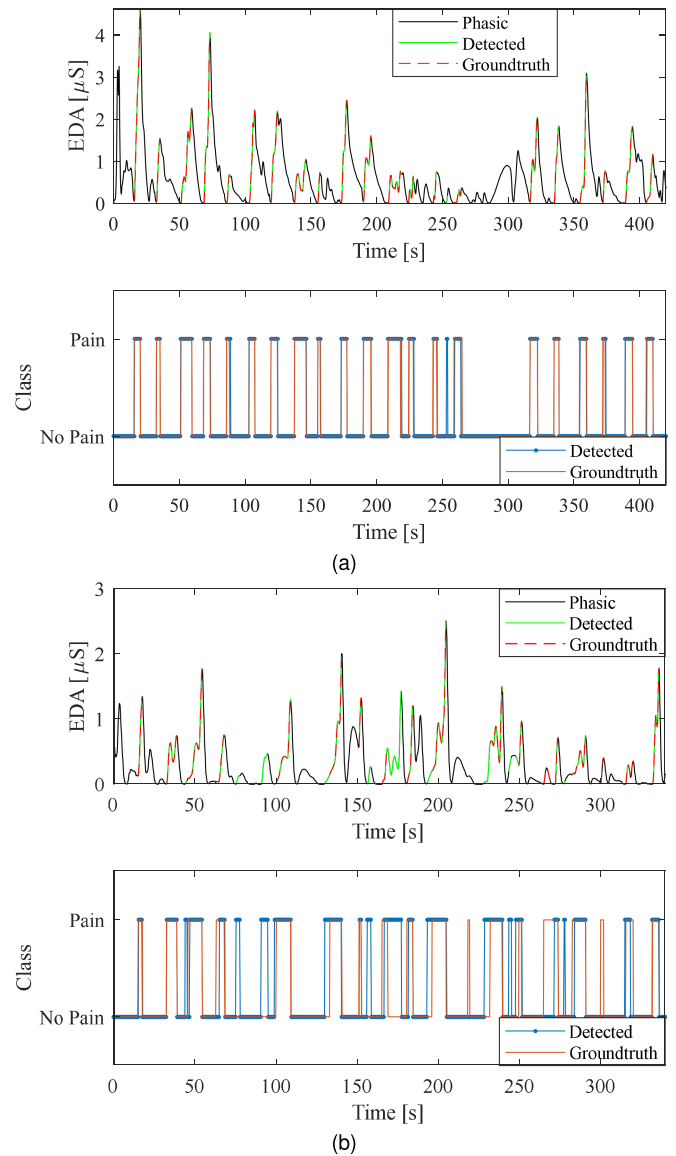


Fig. 3. Continuous pain recognition of EDA signal from two sample subjects using Parallel TCN-SBU-LSTM model using phasic driver and TFS-phEDA as input features. (a) Sample EDA signal in which pain was detected with high accuracy; (b) Sample EDA signal affected by stress and anxiety, reducing the accuracy of the model to detect pain.

a 15-second-long window for each pain stimulus was extracted from the signals, starting from the shifted label time used for labelling the pain curves, as illustrated in Fig. 4. In total, 2,960 windows (740 for each PA) were obtained. All extracted windows contained baseline samples as well.

Fig. 5 shows samples of the pain detection performed by the TCN-SBU-LSTM model on windows corresponding to different pain levels, from a given subject. Note that the y-axis scale of the plots was set to different values to provide a clearer view of the pain detection on each window, as they have different amplitudes resulting from different stimulation levels. When pain response peak was over $0.2 \mu\text{S}$, the networks were able to detect each time step of the sequence, with accurate onset labelling. In contrast, as observed for the PA1 sample, the network only recognized a

TABLE VII
ACCURACY COMPARISON OF OTHER APPROACHES USING THE BioVid Heat Pain Database

Input	Approach	Input	BL vs. PA1	BL vs. PA2	BL vs. PA3	BL vs. PA4	No Pain vs. Pain (PA $\geq$ 1)	No Pain vs. Pain (Continuous detection)
Werner et al. [26]	RF	EDA,ECG,EMG,Video	49.6	60.5	72.0	80.6	-	-
		EDA	55.4	60.2	65.9	73.8	-	-
Wang et al. [27]	Hybrid RNN-ANN	EDA,ECG,EMG	58.5	64.2	75.1	83.3	-	-
Lopez-Martinez et al. [28]	MT-NN	EDA,ECG	51.7	58.6	70.0	82.8	-	-
		EDA	50.0	60.3	69.8	80.0	-	-
Subramaniam et al. [29]	Hybrid CNN-LSTM	EDA,ECG	93.9	87.0	90.8	94.1	97.3	-
		EDA	85.7	74.5	80.8	80.2	92.9	-
Gouverneur et al. [30]	MLP-RF	EDA	59.1	65.1	75.1	84.2	-	-
Our Approach	Parallel TCN-SBU-LSTM	EDA	82.4	86.0	91.8	94.6	88.7	89.8

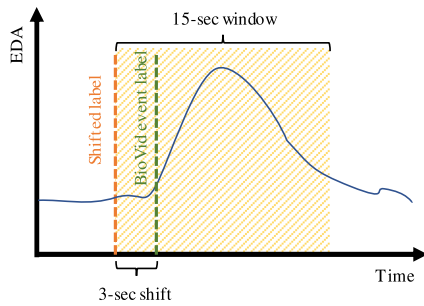


Fig. 4. Window segmentation strategy used for testing the TCN-SBU-LSTM model on the EDA signals from the BioVid database Part C.

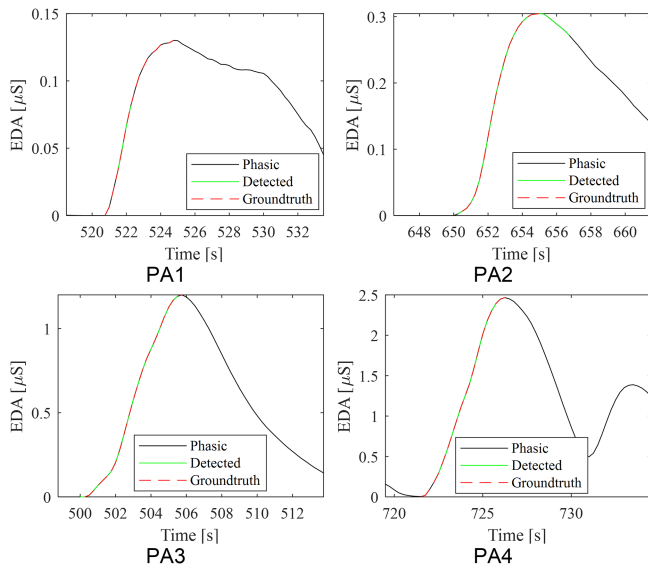


Fig. 5. Examples of pain detection of the TCN-SBU-LSTM model on EDA signal windows from the BioVid Heat Pain Database corresponding to different pain levels.

small segment of the pain response, which was characterized by a peak that is under $0.2 \mu\text{S}$.

As shown in Table VII, our model outperformed the detection at high pain levels, by 1% in PA3 and 0.5% in PA4 from the second-best performance. Only Subramaniam's work [29] tested their performance taking all pain levels as pain, obtaining better results than ours. However, our model maintained a high

performance detecting the pain. Furthermore, none of the compared methods performed continuous pain detection since they only used segmented sections of the signals (Part A of the BioVid database) instead of using the entire signal. In continuous pain detection (by taking the complete signal as input), we obtained 89.8% accuracy.

IV. DISCUSSION

We have examined and trained different deep learning models on various EDA features, such as its components and the TFS-phEDA, for sequence-to-sequence pain/no pain recognition with precise onset and offset time locations of these events. In our previous work, we used machine learning algorithms to do this task, obtaining a maximum accuracy, sensitivity, and specificity of 76%, 72%, and 92%, respectively [20]. In this study, the Parallel TCN-SBU-LSTM model outperformed those results, obtaining a sensitivity score of 82.3%, specificity score of 95.0%, and a global accuracy of 93.1% (77.8% and 0.967 for F1-score and ROC AUC, respectively).

One of the advantages of this type of time series network is that they do not require a fixed length of input signals. However, the performance may change depending on the length of the signal. In our work, the networks were trained with EDA signals of more than 300 seconds. For this reason, to evaluate the effect of the size of the windows of EDA signals on model performance, we tested the Parallel TCN-SBU-LSTM model with different signal lengths. We found that the network was able to successfully recognize the pain responses for signals as small as 15 seconds. Nevertheless, a characteristic misbehavior arose when the pain response occurred before 10 seconds in the signal segment. As shown in Fig. 6(a), the network recognized pain precisely, while in Fig. 6(b), with the same signal shifted 5 seconds backwards, only a portion of the signal was correctly recognized as pain. This characteristic possibly occurred because some of the signals have noise artifacts at the beginning, hence, the network learned to discard activity presented before 10 seconds. The average processing time that the network took to perform pain detection in a 15-second-long signal was 85.6 ms. The processing tests were carried out on a computer with the specifications shown in Table VIII. Therefore, our network can potentially be implemented for real-time applications.

The aim of this study was testing the feasibility of the deep learning techniques for continuous detection of pain using EDA

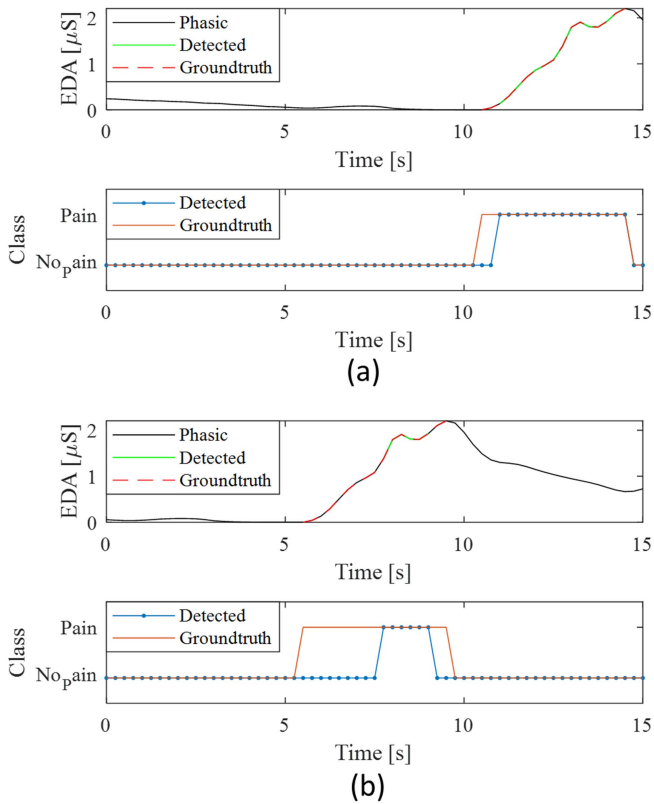


Fig. 6. Performance of the pain detection of the Parallel TCN-SBU-LSTM model is reduced if the event occurred less than 10 seconds within the window. (a) Pain occurred after 10 seconds. (b) Pain occurred before 10 seconds.

TABLE VIII
HARDWARE SPECIFICATIONS

Processor	i7-7700HQ
Generation	7th Generation Intel® Core™
Num. Cores	4
Frequency	2.80 GHz
RAM	32 GB DDR4
GPU	NVIDIA® GeForce® GTX 1070
GPU Memory	8 GB GDDR5
Multiprocessor Count	16
CUDA Cores	2048
Clock Rate	1695 MHz

features, by using shallow architectures. Moreover, for processing signals like EDA, it is not very common to use transfer learning since the dynamics of the signals are not similar to the images that were used for training the transfer learning models. For signals, transfer learning is implemented by training networks on large datasets of data of similar nature to the “smaller” target dataset (images, signals, etc.), and the network is then tuned for the target dataset. Currently, there is no public transfer learning models trained on EDA or similar data. Hence, we decided to avoid networks that required high processing power, allowing our models to be implemented in real-time applications, such as smartphones and wearable devices.

In general, the performance of the deep learning techniques implemented in this study achieved a higher level of

performance in detecting pain-related SCRs in a continuous fashion (sequence-to-sequence), removing the need for sliding windows. Furthermore, the deep learning networks were able to correctly detect segments of EDA variations due to pain in contrast to other confounding factors such as stress and anticipation effects. These results suggest the suitability of deep learning networks to detect pain in realistic scenarios using EDA. Making a comparison between hybrid architectures, the best parallel version in both 1DCNN and TCN versions achieved 76.9% and 77.8% F1-score, respectively, surpassing the residual and series versions by more than 2% F1-score, while having the same depth, parameters, and inputs used (Table V). The possible cause of the worse performance may be due to what the LSTM gets as the input, since the activation features obtained by the CNN affect the training of the recurrent network [53], while the parallel architecture will let the spatial and temporal features be learned without any correlation, and thus, giving to the output section of the network spatial and temporal features, to obtain a better classification.

Examination of various EDA features shows that TFS-phEDA has the potential to be a key physiometer for pain detection, since it achieved the best F1-scores (>76%). When TFS-phEDA was combined with either the phasic driver or the phasic component, classification performance was increased, as shown in Table VI.

One of the limitations of this study is that our database is not considered to be large. This may lead to generalization problems in the pain response recognition. Some studies suggested that the gender [35], [54], or even other demographic factors [55], [56], [57], may influence the pain sensation. For this reason, we performed tests on the BioVid Heat Pain Database with the Parallel TCN-SBU-LSTM model, not only to allow us to make a comparison with other approaches, but to check the robustness of the model. The network outperformed most of the Deep Learning approaches compared for all pain levels and obtained the best performance recognizing high pain levels (BL vs. PA3 and BL vs. PA4), as presented in Table VII. It needs to be remarked that we did not use any of the subjects of the BioVid database to train our network, in addition to the fact that the data acquired by the two databases used in this work employed different protocols (sensors, heat source, place where the pain was inducted, subjects, etc.). This BioVid database was used only as independent test datasets. Hence, the network showed high robustness and generalization in recognizing pain.

Several studies have performed pain level assessment rather than pain detection. In other words, not only the presence of pain, but at what level it is being felt. Thiam et al. evaluated the use of EDA, ECG, and EMG as a single matrix input, as well as separate inputs in different deep learning techniques for pain detection, pain level regression, and multi-class pain classification [58], [59]. Additionally, we have recently used machine learning models for acute pain recognition [20], multi-class pain stimuli sensation, visual analog scale classification [19], and automatic motion artifact detection to improve the pain recognition in EDA signals [60]. In our future study we will test the feasibility of deep learning models to assess the levels of pain.

V. CONCLUSION

We have tested the feasibility of acute continuous pain detection using deep learning techniques, while accurately detecting the onset and offset of pain events. We tested and validated LSTM, 1-dimensional CNN, and different types of hybrid architectures combining those two techniques, for pain detection using EDA signals in healthy subjects. In general, parallel hybrid architectures outperformed other variants tested. The best result was obtained with a parallel combination of a TCN network and an SBU-LSTM network (Parallel TCN-SBU-LSTM), obtaining a global F1-score of 77.8%, almost 1% higher than the second-best result, and with the overall best ROC AUC value of 0.967. Additionally, we found that TFS-phEDA is a highly sensitive feature for pain detection using deep learning models. The best performance was consistently achieved by networks using the TFS-phEDA as part of their input features, with F1-scores of more than 75.7%. Moreover, we tested the Parallel TCN-SBU-LSTM model on the BioVid database, achieving the highest accuracies on BL vs. PA3 classification (91.8%) and BL vs. PA4 classification (94.6%) compared to previously reported approaches for that public database, confirming the generalizability of our model.

REFERENCES

- [1] P. J. Siddall and M. J. Cousins, "Persistent pain as a disease entity: Implications for clinical management," *Anesth. Analg.*, vol. 99, no. 2, pp. 510–520, Aug. 2004, doi: [10.1213/01.ANE.0000133383.17666.3A](#).
- [2] R. K. Portenoy, C. Ugarte, I. Fuller, and G. Haas, "Population-based survey of pain in the United States: Differences among white, African American, and Hispanic subjects," *J. Pain*, vol. 5, no. 6, pp. 317–328, Aug. 2004, doi: [10.1016/j.jpain.2004.05.005](#).
- [3] A. M. Velly and S. Mohit, "Epidemiology of pain and relation to psychiatric disorders," *Prog. Neuropsychopharmacol. Biol. Psychiatry*, vol. 87, pp. 159–167, Dec. 2018, doi: [10.1016/j.pnpbp.2017.05.012](#).
- [4] D. J. Gaskin and P. Richard, "The economic costs of pain in the United States," *J. Pain*, vol. 13, no. 8, pp. 715–724, Aug. 2012, doi: [10.1016/j.jpain.2012.03.009](#).
- [5] G. Chanques et al., "The measurement of pain in intensive care unit: Comparison of 5 self-report intensity scales," vol. 151, pp. 711–721, 2010, doi: [10.1016/j.pain.2010.08.039](#).
- [6] Centers for Disease Prevention, "Drug overdose deaths in the United States continue to increase in 2015," 2015.
- [7] R. A. Rudd, P. Seth, F. David, and L. Scholl, "Increases in drug and opioid-involved overdose deaths—United States, 2010–2015," *Morbidity Mortality Weekly Rep.*, vol. 65, no. 50 & 51, pp. 1445–1452, 2016.
- [8] N. Sehgal, L. Manchikanti, and H. S. Smith, "Prescription opioid abuse in chronic pain: A review of opioid abuse predictors and strategies to curb opioid abuse," *Pain Physician*, vol. 15, no. 3 Suppl, pp. ES67–ES92, Jul. 2012.
- [9] Q. Liu, E. M. Wang, X. J. Yan, and S. L. Chen, "Autonomic functioning in irritable bowel syndrome measured by heart rate variability: A meta-analysis," *J. Dig. Dis.*, vol. 14, no. 12, pp. 638–646, 2013, doi: [10.1111/1751-2980.12092](#).
- [10] H. F. Posada-Quintero and K. H. Chon, "Innovations in electrodermal activity data collection and signal processing: A systematic review," *Sensors*, vol. 20, no. 2, Jan. 2020, Art. no. 479, doi: [10.3390/s20020479](#).
- [11] W. Boucsein et al., "Publication recommendations for electrodermal measurements," *Psychophysiology*, vol. 49, no. 8, pp. 1017–1034, 2012, doi: [10.1111/j.1469-8986.2012.01384.x](#).
- [12] H. F. Posada-Quintero, T. Dimitrov, A. Moutran, S. Park, and K. H. Chon, "Analysis of reproducibility of noninvasive measures of sympathetic autonomic control based on electrodermal activity and heart rate variability," *IEEE Access*, vol. 7, pp. 22523–22531, 2019, doi: [10.1109/ACCESS.2019.2899485](#).
- [13] H. F. Posada-Quintero et al., "Using electrodermal activity to validate multilevel pain stimulation in healthy volunteers evoked by thermal grills," *Amer. J. Physiol.-Regulatory, Integrative Comp. Physiol.*, vol. 319, no. 3, pp. R366–R375, 2020, doi: [10.1152/ajpregu.00102.2020](#).
- [14] H. D. Critchley, "Review: Electrodermal responses: What happens in the brain," *Neuroscientist*, vol. 8, no. 2, pp. 132–142, Apr. 2002, doi: [10.1177/107385840200800209](#).
- [15] T. Aslanidis, V. Grosomanidis, K. Karakoulas, and A. Chatzistiriou, "Electrodermal activity monitoring during painful stimulation in sedated adult intensive care unit patients," *ACTA MEDICA*, vol. 61, no. 2, pp. 47–52, Sep. 2018.
- [16] B. T. Susam et al., "Automated pain assessment using electrodermal activity data and Machine Learning," in *Proc. IEEE 40th Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2018, pp. 372–375, doi: [10.1109/EMBC.2018.8512389](#).
- [17] L. Turner, W. Linden, and C. Marshall, "Electrodermal activity at acupuncture points differentiates patients with current pain from pain-free controls," *Appl. Psychophysiol. Biofeedback*, vol. 38, no. 1, pp. 71–80, Mar. 2013, doi: [10.1007/s10484-013-9209-6](#).
- [18] X. Xu et al., "Towards automated pain detection in children using facial and electrodermal activity," in *Artificial Intelligence in Health*. Cham, Switzerland: Springer, 2019, pp. 181–189, doi: [10.1007/978-3-030-12738-1_13](#).
- [19] H. F. Posada-Quintero, Y. Kong, and K. H. Chon, "Objective pain stimulation intensity and pain sensation assessment using machine learning classification and regression based on electrodermal activity," *Amer. J. Physiol.-Regulatory, Integrative Comp. Physiol.*, vol. 321, no. 2, pp. R186–R196, Aug. 2021, doi: [10.1152/ajpregu.00094.2021](#).
- [20] Y. Kong, H. F. Posada-Quintero, and K. H. Chon, "Real-time high-level acute pain detection using a smartphone and a wrist-worn electrodermal activity sensor," *Sensors*, vol. 21, no. 12, Jan. 2021, Art. no. 3956, doi: [10.3390/s21123956](#).
- [21] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi: [10.1162/neco.1997.9.8.1735](#).
- [22] S. Kiranyaz, O. Avci, O. Abdeljaber, T. Ince, M. Gabbouj, and D. J. Inman, "1D convolutional neural networks and applications: A survey," *Mech. Syst. Signal Process.*, vol. 151, Apr. 2021, Art. no. 107398, doi: [10.1016/j.ymssp.2020.107398](#).
- [23] B. T. Susam et al., "Automated pain assessment in children using electrodermal activity and video data fusion via machine Learning," *IEEE Trans. Biomed. Eng.*, vol. 69, no. 1, pp. 422–431, Jan. 2022, doi: [10.1109/TBME.2021.3096137](#).
- [24] S. A. H. Aqajari et al., "Pain assessment tool with electrodermal activity for postoperative patients: Method validation study," *JMIR mHealth uHealth*, vol. 9, no. 5, May 2021, Art. no. e25258, doi: [10.2196/25258](#).
- [25] E. Kasaeyan Naeini, S. Shahhosseini, A. Subramanian, T. Yin, A. M. Rahmani, and N. Dutt, "An edge-assisted and smart system for real-time pain monitoring," in *Proc. IEEE/ACM Int. Conf. Connected Health: Appl., Syst. Eng. Technol.*, 2019, pp. 47–52, doi: [10.1109/CHASE48038.2019.00023](#).
- [26] P. Werner, A. Al-Hamadi, R. Niese, S. Walter, S. Gruss, and H. C. Traue, "Automatic pain recognition from video and biomedical signals," in *Proc. IEEE 22nd Int. Conf. Pattern Recognit.*, 2014, pp. 4582–4587, doi: [10.1109/ICPR.2014.784](#).
- [27] R. Wang, K. Xu, H. Feng, and W. Chen, "Hybrid RNN-ANN based deep physiological network for pain recognition," in *Proc. IEEE 42nd Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2020, pp. 5584–5587, doi: [10.1109/EMBC44109.2020.9175247](#).
- [28] D. Lopez-Martinez and R. Picard, "Multi-task neural networks for personalized pain recognition from physiological signals," in *Proc. IEEE 7th Int. Conf. Affect. Comput. Intell. Interact. Workshops Demos*, 2017, pp. 181–184, doi: [10.1109/ACIIW.2017.8272611](#).
- [29] S. D. Subramanian and B. Dass, "Automated nociceptive pain assessment using physiological signals and a hybrid Deep learning network," *IEEE Sensors J.*, vol. 21, no. 3, pp. 3335–3343, Feb. 2021, doi: [10.1109/JSEN.2020.3023656](#).
- [30] P. Gouverneur, F. Li, W. M. Adamczyk, T. M. Szikszay, K. Luedtke, and M. Grzegorzek, "Comparison of feature extraction methods for physiological signals for heat-based pain recognition," *Sensors*, vol. 21, no. 14, Jan. 2021, Art. no. 4838, doi: [10.3390/s21144838](#).
- [31] P. Rodriguez et al., "Deep pain: Exploiting long short-term memory networks for facial expression classification," *IEEE Trans. Cybern.*, vol. 52, no. 5, pp. 3314–3324, May 2022, doi: [10.1109/TCYB.2017.2662199](#).
- [32] F. Pouroumran, Y. Lin, and S. Kamathi, "Personalized deep Bi-LSTM RNN based model for pain intensity classification using EDA signal," *Sensors*, vol. 22, no. 21, Jan. 2022, Art. no. 8087, doi: [10.3390/s22118087](#).
- [33] P. Werner, A. Al-Hamadi, K. Limbrecht-Ecklundt, S. Walter, S. Gruss, and H. C. Traue, "Automatic pain assessment with facial activity descriptors," *IEEE Trans. Affect. Comput.*, vol. 8, no. 3, pp. 286–299, Jul.–Sep. 2017, doi: [10.1109/TAFFC.2016.2537327](#).

- [34] E. Othman, P. Werner, F. Saxen, M.-A. Fiedler, and A. Al-Hamadi, "An automatic system for continuous pain intensity monitoring based on analyzing data from uni-, Bi-, and Multi-modality," *Sensors*, vol. 22, no. 13, Jan. 2022, Art. no. 4992, doi: [10.3390/s22134992](https://doi.org/10.3390/s22134992).
- [35] Y. Kong, H. F. Posada-Quintero, and K. H. Chon, "Female-male differences should be considered in physical pain quantification based on electrodermal activity: Preliminary study," in *Proc. IEEE 43rd Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2021, pp. 6941–6944, doi: [10.1109/EMBC46164.2021.9630637](https://doi.org/10.1109/EMBC46164.2021.9630637).
- [36] A. Greco, G. Valenza, A. Lanata, E. P. Scilingo, and L. Citi, "cvxEDA: A Convex optimization approach to electrodermal activity processing," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 4, pp. 797–804, Apr. 2016, doi: [10.1109/TBME.2015.2474131](https://doi.org/10.1109/TBME.2015.2474131).
- [37] P. Welch, "The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Trans. Audio Electroacoustics*, vol. TAE-15, no. 2, pp. 70–73, Jun. 1967, doi: [10.1109/TAU.1967.1161901](https://doi.org/10.1109/TAU.1967.1161901).
- [38] Y. Kong, H. F. Posada-Quintero, and K. H. Chon, "Sensitive physiological indices of pain based on differential characteristics of electrodermal activity," *IEEE Trans. Biomed. Eng.*, vol. 68, no. 10, pp. 3122–3130, Oct. 2021, doi: [10.1109/TBME.2021.3065218](https://doi.org/10.1109/TBME.2021.3065218).
- [39] S. Walter et al., "The biovid heat pain database data for the advancement and systematic validation of an automated pain recognition system," in *Proc. IEEE Int. Conf. Cybern.*, 2013, pp. 128–131, doi: [10.1109/CYB-Conf.2013.6617456](https://doi.org/10.1109/CYB-Conf.2013.6617456).
- [40] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," Apr. 2018, Accessed: Mar. 30, 2022, *arXiv:1803.01271*.
- [41] M. Schuster and K. K. Paliwal, "Bidirectional recurrent neural networks," *IEEE Trans. Signal Process.*, vol. 45, no. 11, pp. 2673–2681, Nov. 1997, doi: [10.1109/78.650093](https://doi.org/10.1109/78.650093).
- [42] S. Siarni-Namini, N. Tavakoli, and A. S. Namin, "The performance of LSTM and BiLSTM in forecasting Time Series," in *Proc. IEEE Int. Conf. Big Data*, 2019, pp. 3285–3292, doi: [10.1109/Big-Data47090.2019.9005997](https://doi.org/10.1109/Big-Data47090.2019.9005997).
- [43] A. Graves, N. Jaitly, and A. Mohamed, "Hybrid speech recognition with Deep bidirectional LSTM," in *Proc. IEEE Workshop Autom. Speech Recognit. Understanding*, 2013, pp. 273–278, doi: [10.1109/ASRU.2013.6707742](https://doi.org/10.1109/ASRU.2013.6707742).
- [44] R. L. Abduljabbar, H. Dia, and P.-W. Tsai, "Unidirectional and bidirectional LSTM models for short-term traffic prediction," *J. Adv. Transp.*, vol. 2021, Mar. 2021, Art. no. e5589075, doi: [10.1155/2021/5589075](https://doi.org/10.1155/2021/5589075).
- [45] Z. Cui, R. Ke, Z. Pu, and Y. Wang, "Deep bidirectional and unidirectional LSTM recurrent neural network for network-wide traffic speed prediction," Nov. 2019, Accessed: Apr. 21, 2022, *arXiv:180102143*.
- [46] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015, doi: [10.1038/nature14539](https://doi.org/10.1038/nature14539).
- [47] S. Kiranyaz, T. Ince, R. Hamila, and M. Gabbouj, "Convolutional Neural networks for patient-specific ECG classification," in *Proc. IEEE 37th Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2015, pp. 2608–2611, doi: [10.1109/EMBC.2015.7318926](https://doi.org/10.1109/EMBC.2015.7318926).
- [48] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," Jul. 2016, Accessed: Mar. 31, 2022, *arXiv:160706450*.
- [49] X. Liu, J. Jia, and R. Zhang, "Automatic detection of epilepsy EEG based on CNN-LSTM Network combination model," in *Proc. 4th Int. Conf. Comput. Sci. Artif. Intell.*, 2020, pp. 225–232, doi: [10.1145/3445815.3445852](https://doi.org/10.1145/3445815.3445852).
- [50] X. Ma, J. Zhang, B. Du, C. Ding, and L. Sun, "Parallel architecture of convolutional Bi-directional LSTM neural networks for network-wide metro ridership prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 6, pp. 2278–2288, Jun. 2019, doi: [10.1109/TITS.2018.2867042](https://doi.org/10.1109/TITS.2018.2867042).
- [51] M. Xu, Z. Yin, M. Wu, Z. Wu, Y. Zhao, and Z. Gao, "Spectrum sensing based on parallel CNN-LSTM network," in *Proc. IEEE 91st Veh. Technol. Conf.*, 2020, pp. 1–5, doi: [10.1109/ACCESS.2020.9129229](https://doi.org/10.1109/ACCESS.2020.9129229).
- [52] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 9260–9269, doi: [10.1109/CVPR.2019.00949](https://doi.org/10.1109/CVPR.2019.00949).
- [53] B. Farsi, M. Amayri, N. Bouguila, and U. Eicker, "On short-term load forecasting using machine learning techniques and a novel parallel deep LSTM-CNN approach," *IEEE Access*, vol. 9, pp. 31191–31212, 2021, doi: [10.1109/ACCESS.2021.3060290](https://doi.org/10.1109/ACCESS.2021.3060290).
- [54] D. S. Bari, "Gender differences in tonic and phasic electrodermal activity components," *Sci. J. Univ. Zakho*, vol. 8, no. 1, Mar. 2020, Art. no. 1, doi: [10.25271/sjuoz.2020.8.1.670](https://doi.org/10.25271/sjuoz.2020.8.1.670).
- [55] K. M. Woodrow, G. D. Friedman, A. B. Siegelau, and M. F. Collen, "Pain tolerance: Differences according to age, sex and race," *Psychosom. Med.*, vol. 34, no. 6, pp. 548–556, Nov. 1972.
- [56] I. Kvachadze, M. G. Tsagareli, and Z. Dumbadze, "An overview of ethnic and gender differences in pain sensation," *Georgian Med. News*, vol. 22, no. 238, pp. 102–108, Jan. 2015.
- [57] T. Pillay, H. A. van Zyl, and D. Blackbeard, "Chronic pain perception and cultural experience," *Procedia-Soc. Behav. Sci.*, vol. 113, pp. 151–160, Feb. 2014, doi: [10.1016/j.sbspro.2014.01.022](https://doi.org/10.1016/j.sbspro.2014.01.022).
- [58] P. Thiam, P. Bellmann, H. A. Kestler, and F. Schwenker, "Exploring deep physiological models for nociceptive pain recognition," *Sensors*, vol. 19, no. 20, Jan. 2019, Art. no. 4503, doi: [10.3390/s19204503](https://doi.org/10.3390/s19204503).
- [59] P. Thiam, H. Hihn, D. A. Braun, H. A. Kestler, and F. Schwenker, "Multi-modal pain intensity assessment based on physiological signals: A deep learning perspective," *Front. Physiol.*, vol. 12, Sep. 2021, Art. no. 720464, doi: [10.3389/fphys.2021.720464](https://doi.org/10.3389/fphys.2021.720464).
- [60] M.-B. Hossain, H. F. Posada-Quintero, Y. Kong, R. McNaboe, and K. H. Chon, "Automatic motion artifact detection in electrodermal activity data using machine learning," *Biomed. Signal Process. Control*, vol. 74, Apr. 2022, Art. no. 103483, doi: [10.1016/j.bspc.2022.103483](https://doi.org/10.1016/j.bspc.2022.103483).